

Infix to postfix

20 October 2023 10:14

- 1) start scanning from left to right
- 2) If (current element is an operand)
Append it to the postfix expression
- 3) Else if (current element is opening bracket '(')
Push it onto the stack
- 4) Else if current element is closing bracket ')'
 - Pop elements from the stack and append them to postfix exp till its corresponding opening brackets does not occur
 - Pop opening brackets from the stack and discard both the brackets
- Else
 - // If (current element is an operator)
 - While (stack is not empty && priority of topmost element >= priority of current element)
 - { Pop element from the stack and append it to postfix expression
 - }
 - Push current element onto the stack.
- 5) Repeat the above steps till end of infix expression
- 6) Pop all the remaining elements from the stack one by one and append them to postfix expression

Infix expression :

5+9-4*(8-6/2)+1\$(7-3)

Current element :

Postfix expression: 59+4862/-*-173-\$+

stack



top

$5+2 \rightarrow \text{infix.}$
 operands $\rightarrow 5, 2.$
 operator $\rightarrow +.$
 $\downarrow$
 $\begin{cases} +52 \rightarrow \text{prefix.} \\ 52+ \rightarrow \text{postfix.} \end{cases}$

Conversion $\rightarrow$ stack } algo
 output $\rightarrow$ stack.

✓ 5+9-4*(8-6/2)+1\$(7-3)

()
 ^ OR \$
 * / %
 + -

⑥ ⑦ ⑤ ② ① ⑧ ④ ③
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)
 5+9-4*(8-6/2)+1\$(7-3)

operand
 operator
 (
)

$$59 + - 4862 / - * + 173 - \$$$

$$59 + 4862 / - * - + 173 - \$$$

$$59 + 4862 / - * - 173 - \$ +$$

→ postfix exp.

59+4862/-*-173-\$+